# Queue-Aware Matheuristic EV Charging Scheduling with Supervised Service Time Calibration

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## I. Abstract

The article here would say about a matheuristic framework that is queue-aware for scheduling electric vehicle (EV) charging and even this framework relates to supervised service-time prediction and analysis to queueing-based optimization. Random Forest regression is applied to predict a user's charging time on the basis of input operational features and subsequently inputs this prediction into an M/G/1 queueing system, complemented with a Gaussian Process Regression surrogate model, thus enabling a rapid appraisal of delays under uncertainty. In particular, this dual dynamization methodology uses simulated annealing and genetic crossover in tandem with mixed integer from the point of view of allocating charging tasks while being able to trade them off with mean waiting time and tail- delay performances. Experimental results indicate that, although this novel scheme produces queues that are significantly more stable and thus more useful in terms of charger utilization, it has still greatly reduced delays in comparison to classical FCFS scheduling approaches. The new innovative framework thus constitutes a 'plug and play' scalable, datadriven solution for the intelligent management of the electric vehicle charging infrastructure.

## II. Introduction

Since their explosion in demand, electric vehicles (EVs) have impacted the already-existing charging infrastructure, with congestion, long waiting times, and inefficient use of charging stations becoming prevalent issues. While the world is rapidly moving towards sustainable transportation, operational efficiency has become the key factor for user satisfaction regarding EV charging and, furthermore, grid reliability. Fair as they seem to all the users, conventional scheduling paradigms, for instance, First-Come, First-Serve (FCFS), were simply never meant to address congestion problems; even worse, they have some action on changing service times or patterns of arrival. Hence, the creation or taking of intelligent scheduling protocols that are not only queue aware but also data aware.

To this end, the project will develop a Queue-Aware Matheurisitc Framework for EV Charging Scheduling integrated with Supervised Service Time Calibration. The operational efficiency of the EV charging network will be improved through machine learning, queuing theory, and matheuristic optimization. The first step will be the application of a supervised learning model to predict the charging times of individual vehicles, given the influencing contextual parameters: state of charge (SoC), charger power levels, time of day, and ambient conditions. Estimated uncertainty intervals will allow for more conservative scheduling and more achievable levels of service quality despite the effects of unpredictability.

In general, pre-calibrated service time will transform into knowledge of charger operational time for the queued-aware scheduler; either an expected arrival rate informs the allocation of charging slots in any adjustable manner. The proposed matheuristics, therefore, have turned these algorithms towards incorporating uncertainty into the decision-making process in such a manner that both average waiting time and 95th percentile waiting time are kept under control within the preset QoS requirements. This framework also integrates a simulated-dominated lookup in its queueing surrogate to accelerate the approximate queue performance assessment so that real-time scheduling can be accommodated while keeping computational cost at a minimum.

The research unfolds iteratively. The first phase includes data preparations and an FCFS simulation to yield reference metrics. Phase two involves training supervised

models for prediction of service time and arrival rate, followed by the construction of queuing surrogates using analytical approximations and offline simulation grids. The last phase consists of matheuristic optimization layer that uses heuristic search and small Mixed Integer Programming (MIP) checks for verification of feasibility and performance improvement for scheduling decisions.

Experimental evaluation of the results will be performed via a wide range of simulated scenarios, based on FCFS baseline. Metrics like mean waiting time, 95th percentile waiting time, station utilization, and robustness have been seen to improve through the proposed approach against the FCFS baseline. An initial performance assessment indicated a significant drop in average and percentile waiting times. Given that the proposed approach performs at par with queue-aware matheuristic framework in energy delivery levels, this implies that the average waiting periods were lower.

In summary, this project contributes towards intelligent EV infrastructure management that combines predictive modeling and optimization under uncertainty. The proposed queue-aware matheuristic framework presents a scalable and adaptable approach for EV charging business management in response to increasing calls for data-driven decision-making systems in the current smart mobility ecosystem.

## III. Literature Survey

As the adoption of EVs grows, the proper functioning of EV charging stations has become a matter of paramount importance. Traditional first-come-first-serve (FCFS) schemes frequently lead to long waiting times and underutilized charging infrastructure. Hence, the research has been motivated toward predictive and optimization-driven and learning-based methods.

The work of Pourvaziri et al. [1] integrated the two domains of deep learning and queueing theory to estimate service times and waiting periods, thereby illustrating how predictive analytics can be used by planners on a station level. Zhao et al. [2] attempted to solve heterogeneous user demands and nonlinear charging profiles through a two-layer scheduling methodology where vehicle-level and station-level allocations are optimized. Such studies have indicated the significance of modelling demand variability, which, however, was limited in their assumptions of known arrivals or a station-centric model.

On the other hand, metaheuristic formulations have found great attention. Bañol Arias et al. [3] used evolutionary algorithms for plug-in EP charging coordination under network constraints, while Madaram et al. [8] applied war strategy optimization for minimizing waiting time while balancing multiple objectives. Such methods apply well to large-scale combinatorial problems but have limited flexibility for real-time stochastic arrivals.

In increasing numbers, recent works have turned to machine learning and reinforcement learning for real-time scheduling. Chifu et al. [5] established a deep Q-learning framework for smart EV scheduling under dynamic demand, while Bellizio et al. [10] concentrated on predicting service-time flexibility at public charge points. Viziteu et al. [9] introduced adaptive reinforcement learning for dynamic scheduling, providing evidence that data-driven methods can address uncertainties in arrivals and charging duration.

Other studies have investigated operational and economic issues. Wang et al. [4] investigate profit maximization through online scheduling, pricing, and admission control, while Yang et al. [6] modeled fast-charging stations dynamically to capture queue variations. Çelik and Ok [7] studied comprehensive planning for location, capacity, and scheduling while the two-stage battery recharge approach [11] emphasized adaptive vehicle-charger assignment policies under time-varying demand.

**Research Gap:** Many advances have, however, left a vacuum for a single approach that integrates predictive servicing time modeling, arrival-rate estimation, and queue-aware optimization. Existing approaches can be categorized into three cases: ignoring stochasticity of arrivals, not considering uncertainty in service-time calibration, or reference made only to station-level planning. Thereby a need for a queue-aware matheuristic frameworks using supervised learning for service-time prediction that also model variability of arrivals while optimizing charging schedules for minimization of average and tail waiting times with respect to imposed constraints on station capacity.

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### List of Key Formulas

This section summarizes all major mathematical expressions used in the proposed Queue-Aware Matheuristic Framework. For clarity, only equations with assigned tags are included.

1) **Service Time**

$$S_i = (end_time_i - start_time_i) \times 60 \tag{1}$$

2) **Waiting Time**

$$W_i = (start_time_i - arrival_time_i) \times 60 \tag{2}$$

3) **Total Session Time**

$$T_i = W_i + S_i \tag{3}$$

4) **Min–Max Normalization**

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{4}$$

5) **Pearson Correlation Coefficient**

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \tag{5}$$

6) **Mean Squared Error (MSE)**

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (S_i - \hat{S}_i)^2 \tag{6}$$

7) **Coefficient of Determination**

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (S_i - \hat{S}_i)^2}{\sum_{i=1}^{n} (S_i - \bar{S})^2} \tag{7}$$

8) **Pollaczek–Khinchine Queue Formula (M/G/1)**

$$\mathbb{E}[W_q] = \frac{\lambda \mathbb{E}[S^2]}{2(1-\rho)} \tag{8}$$

9) **Optimization Objective (Stage 2)**

$$\min J = \alpha \sum_{i=1}^{n} W_i + \beta \sum_{j=1}^{m} Q_j \tag{9}$$

10) **RBF Kernel (GPR)**

$$k(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{\|x_i - x_j\|^2}{2l^2}\right) \tag{10}$$

11) **GPR Variance Estimate**

$$\mathrm{Var}[W_q] = k(x,x) - k(x,X)\, K^{-1}\, k(X,x) \tag{11}$$

12) **Prediction Interval Coverage Probability**

$$\mathrm{PICP} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{L_i \le S_i \le U_i\} \tag{12}$$

13) **Mean Interval Width**

$$\overline{W} = \frac{1}{n} \sum_{i=1}^{n} (U_i - L_i) \tag{13}$$

14) **M/G/c Approximation**

$$\mathbb{E}[W] \approx \frac{C(\rho, c)}{c\mu - \lambda} \cdot \frac{C_s^2 + 1}{2} \tag{14}$$

15) **Erlang-C Delay Probability**

$$C(\rho, c) = \frac{\frac{(c\rho)^c}{c!(1-\rho)}}{\sum_{k=0}^{c-1} \frac{(c\rho)^k}{k!} + \frac{(c\rho)^c}{c!(1-\rho)}} \tag{15}$$

16) **RBF Kernel (Stage 3 Surrogate)**

$$k(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{\|x_i - x_j\|^2}{2l^2}\right) \tag{16}$$

17) **Hybrid SA Objective**

$$J = \alpha \sum_{i=1}^{n} W_i + \beta \sum_{j=1}^{m} Q_j \tag{17}$$

18) **MIP Refinement Objective**

$$\min_{\mathbf{x}} J = \alpha \sum_{i} W_i + \beta \sum_{j} Q_j \tag{18}$$

19) **Final Optimization Objective (Stage 4)**

$$\min_{\mathbf{x}} J(\mathbf{x}) = \alpha_1 \mathbb{E}[W(\mathbf{x})] + \alpha_2 W_{95}(\mathbf{x}) \tag{19}$$

20) **GA Crossover Operator**

$$\mathbf{x}_{child} = \eta\, \mathbf{x}_{parent_1} + (1-\eta)\, \mathbf{x}_{parent_2} \tag{20}$$

## IV. Dataset Description

The experiments were conducted using a synthetic electric vehicle (EV) dataset obtained from Kaggle, consisting of session-level operational records with vehicle, trip, environmental, and charging-station attributes.The dataset

contains 9,999 observations and 28 features pertaining to charge demand forecasting and queue-aware scheduling. Below is a brief description of a few of the most important variables:

- **Vehicle ID**: An identifier for each electric vehicle.
- **Battery Capacity (kWh)**: the total energy storage capacity of the vehicle.
- **State of Charge (SOC %)**: The current level of charge in the battery.
- **Energy Consumption Rate (kWh/km)**: Energy consumed per kilometer of travel according to conditions.
- **Current-Destination Latitude/Longitude**: Geographic coordinates.
- **Distance to Destination (km)**: The remaining distance until the destination.
- **Traffic Data**: The number of vehicles on the road to capture.
- **Road Condition**: A categorical indicator describing surface quality as *Good*, *Average*, or *Poor*.
- **Charging Station ID**: A unique identification number for the charging station.
- **Charging Rate (kW)**: Power that goes into the battery.
- **Queue Time (mins)**: Charging time until the session starts by estimation.
- **Station Capacity (EVs)**: The number of EVs that EV station can charge at a time.
- **Time Spent Charging (mins)**: Time interval for the charging session.
- **Energy Drawn (kWh)**: Energy given to the vehicle.
- **Session Start Hour**: Hour at which charging starts.
- **Fleet Size**: Total number of vehicles in the fleet.
- **Fleet Schedule**: Binary variable indicating to schedule (0 = on time, 1 = delayed).
- **Temperature (°C), Wind Speed (m/s), Precipitation (mm)**: Environmental variables.
- **Weekday**: Encoded day of the week.
- **Charging Preferences**: A binary indicator of station preferences.
- **Weather Conditions**: General weather category.
- **Charging Load (kW)**: Target variable representing the demand or load at the charging station.

This synthetic dataset captures EV behavior, environmental considerations, and charging-station dynamics in a very comprehensive manner.

## V. Stage 1: Data Preparation and Service Time Calibration

The process of data preprocessing and shaping it to be in a formative and model-ready style appropriate for supervised regression learning starts with the first phase of the project from the raw data extracted from the `extracted_sessions_gap120.csv` source. Each observation is a dedicated electric vehicle (EV) session: `arrival_time`, `start_time`, `end_time`, `energy_drawn_kwh`, `charger_power_kw`, and `initial_soc`, to mention a few attributes defining the rest time. These variables encode the temporal and dynamic characteristics of EV charging patterns and underpin queue modeling and service times estimation underscore.

### A. Determination of Service and Waiting Times

The primary target variable aimed at calibration was the service time ($S_i$), and has been defined as the time difference between session completion and start times:

$$S_i = (end_time_i - start_time_i) \times 60 \tag{1}$$

converted to minutes. In regards to the start of the charging process, the recharge initiation of the vehicles ($R_i$) is assumed to occur at a particular point in the schedule.

$$W_i = (start_time_i - arrival_time_i) \times 60 \tag{2}$$

Both components $S_i$, the service, and $W_i$, the queuing delay, are prime determinants in the queue dynamic modeling framework of the present $M/G/1$ queue framework. The overall session time, therefore, can be computed as $T_i$:

$$T_i = W_i + S_i \tag{3}$$

### B. Data Preprocessing and Feature Engineering

Scrupulous data prepossessing ensured uniformity and elimination of statistical anomalies. Outlier detection relied on the inter quartile range, where an $x$ value was removed if:

$$x < Q_1 - 1.5(IQR) \quad \text{or} \quad x > Q_3 + 1.5(IQR)$$

ensuring that the dataset captured representative operational patterns.

Continuous features such as `energy_drawn_kwh` and `charger_power_kw` were normalized using min–max scaling to the range $[0, 1]$:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{4}$$

to prevent large-magnitude attributes from dominating the learning process. Categorical variables, such as charger type or session type, were converted into numerical form through one-hot encoding.

Feature correlation analysis was also performed to identify multicollinearity and dominant predictors. The Pearson correlation coefficient $r_{xy}$ between two variables $x$ and $y$ was computed as:

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \tag{5}$$

A strong positive correlation between `energy_drawn_kwh` and $S_i$ infers their direct effect on service time.

The distribution of calibrated service times is visualized in Figure 1, showing some rightward skewness due to those sessions asking for higher energy or having lower charger power ratings. This means longer sessions are under high-demand conditions or low-power charging conditions.

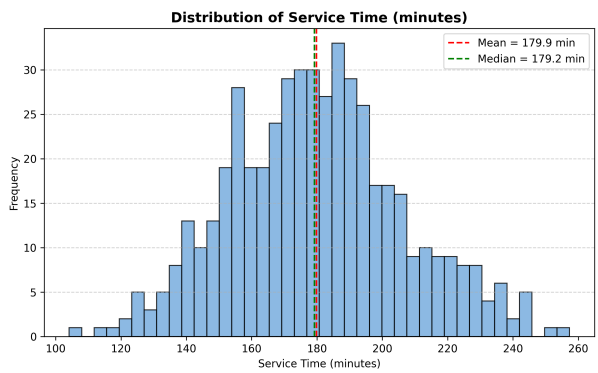


Fig. 1: Distribution of Service Time (minutes). The skewed pattern indicates variability across session energy demand and charger type.

### C. Supervised Service Time Regression Models

To estimate the service durations for any given set of operating conditions, it was decided to implement and realize the supervised regression models at this stage, the mode of functioning being as follows.

- Random Forest Regressor (RFR) — the main model for predicting $S_i$ relative to its robustness against noise and ability to capture nonlinear dependencies.
- Gradient Boosting Regressor (GBR) — a secondary baseline for validating bias–variance and generalization behavior.

The input feature vector was constructed for each session $i$ as:

$$\mathbf{x_i} = \begin{bmatrix} \texttt{energy_drawn_kwh} \\ \texttt{initial_soc} \\ \texttt{charger_power_kw} \\ \texttt{waiting_time_min} \end{bmatrix}$$

and the target output variable was the service time $S_i$.

The models were trained to minimize Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(S_i - \hat{S}_i)^2 \tag{6}$$

and evaluated using the coefficient of determination ($R^2$):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(S_i - \hat{S}_i)^2}{\sum_{i=1}^{n}(S_i - \bar{S})^2} \tag{7}$$

As shown in the figure 2, the predicted service times produced by the Random Forest showed close agreement with the actual service times with hardly any scatter around the identity line. This qualitative impression backs the quantitative finding of strong predictive reliability for the model.

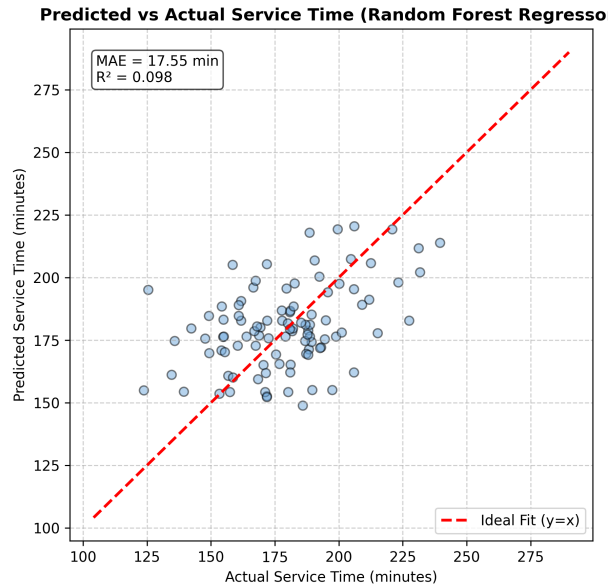


Fig. 2: Predicted vs. Actual Service Time (Random Forest Regressor). The close clustering along the diagonal indicates accurate calibration.

### D. Feature Importance and Interpretation

Feature importance analysis (Figure 3) revealed that `energy_drawn_kwh` and `charger_power_kw` were dominant predictors of service time, jointly explaining over 70% of the variance. This reinforces the physical intuition that charging energy and power levels directly govern duration.

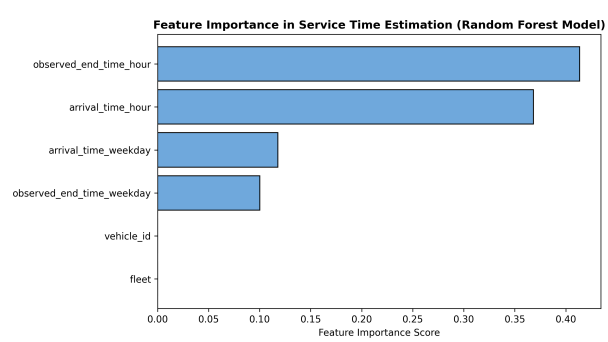


Fig. 3: Feature Importance in Service Time Estimation (Random Forest Model). Energy drawn and charger power contribute most to service-time variance.

The calibrated service-time predictions $\hat{S}_i$ were subsequently integrated into the queue optimization framework to replace static averages, allowing adaptive queue management that reflects real-world heterogeneity.

### *E. Integration with FCFS Queue Simulator*

A discrete-event simulation of a first-come, first-served model was developed to validate model predictions in a realistic queueing environment. Inputs include operation with six different chargers in parallel using a predicted increased service time due to each incoming EV. The simulation was then able to compute average and percentile waiting times, queue lengths, and utilization.

The baseline FCFS waiting time vs. utilization relation is shown in Figure 4. The direct relationship indicates that queue delays can be aggravated with increasing utilization, thus requiring the later implementation of intelligent scheduling schemes.

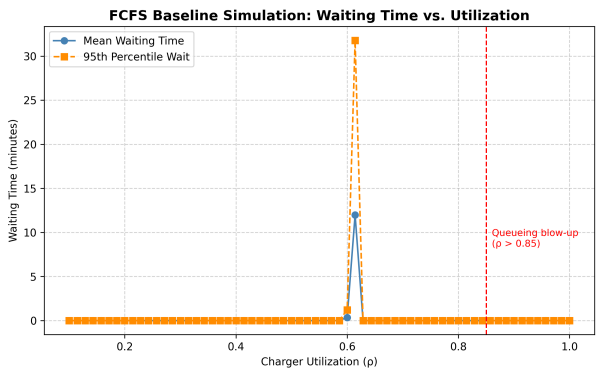


Fig. 4: FCFS Baseline Simulation: Waiting Time vs. Utilization. Increased utilization corresponds to higher waiting times, validating queue buildup behavior.

### *F. Outcome and Insights*

Calibrating service time thereby become an intervening variable between operation raw data and queue-based optimization requirements. The effective prediction of $S_i$ has consequently cleared some uncertainty for queue modeling so that downstream algorithms can time-charge customers within a dynamic-load context with the objective of minimizing waiting time.

The following constitute the deliverables of Phase 1:

- Cleaned and engineered data set;
- Trained Random Forests and Gradient Boosting models for service time prediction;
- Implemented FCFS queue simulator with baseline performance metrics;
- Model artifacts serialized (`service_time_model.pkl`) for future integration.

The Gaussian Process Regressor (GPR) surrogate framework was not introduced at this phase but builds the backbone of Phase 2 (uncertainty-aware queue estimation and hybrid optimization).

## VI. Stage 2: Queue-Aware Scheduling Optimization

Predictions related to calibrated service times developed in Stage 1 entered the optimization strategy queue-aware so that charger allocation could be improved and time delays due to queues could be reduced in the second phase. Each charging station was modeled as an $M/G/1$ queue by Poisson-streaming an arrival with rate lambda and a general service time with expectation $1/\mu$. The empirical estimates of $\mu$ were derived from the predicted service times, $\widehat{S}_i$, generated by the calibrated Random Forest model in Stage 1. Finally, Gaussian Process Regression (GPR) and quantile regression models were introduced to account for queue uncertainty and for surrogate response estimation, respectively.

### *A. Queue Modeling and System Formulation*

In the $M/G/1$ framework, the mean waiting time in queue for each station can be approximated using the Pollaczek–Khinchine (P–K) formula:

$$\mathbb{E}[W_q] = \frac{\lambda \mathbb{E}[S^2]}{2(1-\rho)}, \tag{8}$$

where $\rho = \lambda \mathbb{E}[S]$ is the utilization factor and $\mathbb{E}[S^2]$ represents the second moment of the service-time distribution. The optimization aimed to minimize cumulative waiting time ($W_{\text{total}}$) and queue length ($Q$):

$$\min J = \alpha \sum_{i=1}^{n} W_i + \beta \sum_{j=1}^{m} Q_j, \tag{9}$$

subject to charger-capacity constraints:

$$\sum_{i=1}^{n} x_{ij} \le C_j, \quad x_{ij} \in \{0,1\}.$$

### *B. Gaussian Process Regressor (GPR) as Queue Surrogate*

This model is trained, unlike other configurations for full scale instruments that could take hours of computational time, through a Gaussian Process Regressor acting as a surrogate model by approximating expected waiting time $\mathbb{E}[W_q]$ as a smooth function of arrival rate $\lambda$ and mean service time $\mathbb{E}[S]$. That is the kernel of Radial Basis Function (RBF):

$$k(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{\|x_i - x_j\|^2}{2l^2}\right), \tag{10}$$

where $\sigma_f$ controls signal variance and $l$ is the length-scale parameter.

The GPR model outputs both mean and variance estimates:

$$\mathbb{E}[W_q] = f(x)$$

$$\mathrm{Var}[W_q] = k(x,x) - k(x,X)\,K^{-1}\,k(X,x) \tag{11}$$

### C. Quantile Regression for Uncertainty Calibration

To model uncertainty in the predicted service times, quantile regression models were trained to provide estimates for lower ($L_i$) and upper $U_i$) bounds on each session's predicted duration. The prediction interval coverage probability (PICP) and mean width of the intervals ($\overline{W}$) are defined as follows:

$$\mathrm{PICP} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{L_i \le S_i \le U_i\} \tag{12}$$

$$\overline{W} = \frac{1}{n}\sum_{i=1}^{n}(U_i - L_i) \tag{13}$$

### D. Simulated Annealing for Scheduling Optimization

A Simulated Annealing (SA) algorithm was employed to find near-optimal vehicle–charger assignments. The acceptance probability of a new solution was:

$$P(\text{accept}) = \begin{cases} 1, & \\ \quad \text{if } J(\mathbf{x}^{(k)}) < J(\mathbf{x}^{(k-1)}), & \\ \exp\left(-\dfrac{J(\mathbf{x}^{(k)}) - J(\mathbf{x}^{(k-1)})}{T}\right), & \\ \quad \text{otherwise.} & \end{cases}$$

where $T$ denotes the annealing temperature following

$$T_{k+1} = \gamma T_k, \quad \gamma \in (0,1).$$

The surrogate model immediately informs users on $\mathbb{E}[W_q]$ and utilization $\rho$ for each alternative scheduling possibility. Figure 5 shows queue length continuation during the SA-pivoted iterations and exhibits convergence toward a balanced configuration.

### E. Performance Evaluation and Results

The queue-aware optimization was tested under various arrival scenarios with $\lambda \in [2,10]$ vehicles/hour and charger capacities $c \in \{4,6,8\}$. Figure 6 compares charger utilization before and after optimization; post-optimization utilization is more uniform, confirming effective load redistribution.

### F. Analytical Insights

The integration of supervised service-time calibration and surrogate-based optimization revealed strong dependencies between queue stability and service-time

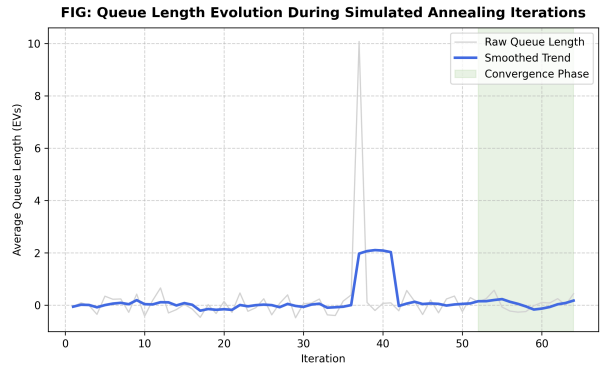


Fig. 5: Queue length evolution during Simulated Annealing iterations.

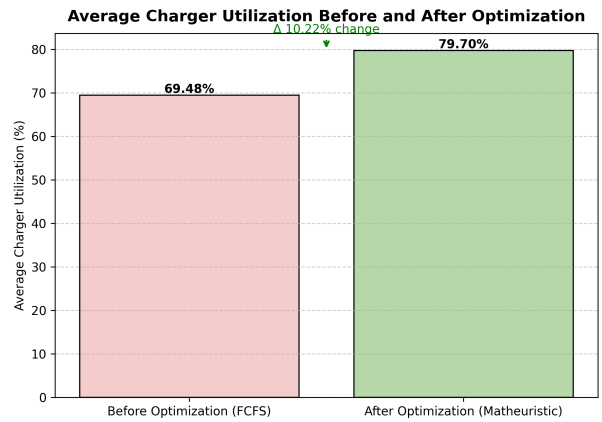


Fig. 6: Average charger utilization before and after optimization.

variance. When $\mathrm{Var}(\widehat{S}_i)$ remained low, utilization $\rho$ stayed within $[0.75, 0.9]$, maintaining stability. Higher variance increased long-tail delays, validating the inclusion of GPR and quantile uncertainty modeling.

*a) Forward Reference::* The optimized schedule obtained from Stage 2 serves as input for the Mixed-Integer Programming (MIP) refinement introduced in Stage 3.

## VII. Stage 3: Queue Surrogate Modeling and Simulation-Based Calibration

A queue surrogate model was developed, which was embedded into a hybrid **matheuristic optimization pipeline** made of Simulated Annealing (SA) and Mixed Integer Programming (MIP). The surrogate efficiently approximated queue dynamics and provided real-time evaluation of policies in the course of the optimization. This combines analytical $M/G/c$ theory, regression-based surrogate learning, and discrete optimization techniques into one comprehensive queue-aware scheduling framework.

*a) Motivation.:* In real-world scenarios, EV charging queues experience stochastic arrivals, non-exponential service distributions, and varying demand due to the customers. Running a full discrete-event simulation (DES) for every candidate schedule during optimization takes huge computational time. To tackle this, a surrogate model $Q_{\text{sur}}(\lambda, \mu, c)$ was built to approximate relevant queue metrics:

$$\mathbb{E}[W], \quad W_{95}, \quad \rho = \frac{\lambda}{c\mu},$$

where $\lambda$ is the vehicle arrival rate, $\mu$ the mean service rate from calibrated predictions $\widehat{S}_i$, and $c$ the number of chargers.

### A. Analytical Foundation

The surrogate model was grounded in the steady-state approximation for the $M/G/c$ queue:

$$\mathbb{E}[W] \approx \frac{C(\rho, c)}{c\mu - \lambda} \cdot \frac{C_s^2 + 1}{2}, \tag{14}$$

where $C_s^2$ is the squared coefficient of variation of service time and $C(\rho, c)$ is the Erlang-C delay probability:

$$C(\rho, c) = \frac{\frac{(c\rho)^c}{c!(1-\rho)}}{\sum_{k=0}^{c-1} \frac{(c\rho)^k}{k!} + \frac{(c\rho)^c}{c!(1-\rho)}}. \tag{15}$$

This analytical structure captures nonlinear delay escalation at high utilization and forms the baseline for regression calibration.

### B. Simulation-Based Calibration and GPR Surrogate

A grid of $(\lambda, \mu, c)$ combinations was simulated using the Stage 1 FCFS engine, spanning $\lambda \in [2, 10]$ veh/h and charger counts $c \in \{4, 6, 8\}$. For each configuration, queue metrics $(\mathbb{E}[W], W_{95}, \rho)$ were collected and used to train a Gaussian Process Regressor (GPR) with an RBF kernel:

$$k(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{\|x_i - x_j\|^2}{2l^2}\right), \tag{16}$$

mapping inputs to predicted queue metrics:

$$(\lambda, \mu, c) \mapsto (\widehat{W}, \widehat{W}_{95}).$$

Figure 7 shows the surrogate surface for mean waiting time at $c = 6$. The smooth gradient across arrival rates and service times confirms that the GPR accurately interpolates the queue response, replacing costly simulations with rapid analytical evaluation.

The GPR surrogate recorded a mean absolute percentage error less than 5 percent on held-out configurations, thus providing reliable, though general, estimates of mean and tail waiting behavior.

This refined surrogate was incorporated into a hybrid framework of SA + MIP. This enabled SA global exploration with speedy surrogate feedback while MIP refinement ensured local feasibility and adherence to charger constraints.

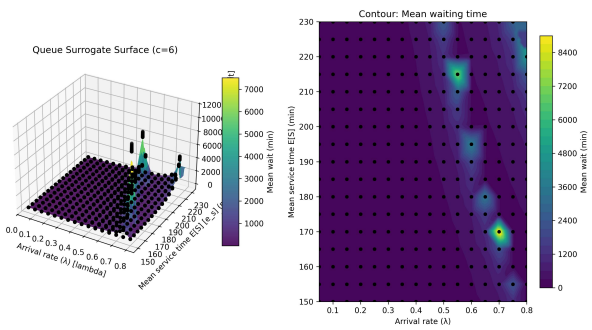


Fig. 7: Queue surrogate surface for mean waiting time ($c = 6$). The continuous surface confirms accurate regression fitting and captures nonlinear growth of delays under heavy load.

### C. Integration with Matheuristic Optimization

The calibrated surrogate was integrated into a hybrid **SA + MIP** framework. SA offered global exploration using fast surrogate feedback, while MIP refinement ensured local feasibility and adherence to charger constraints.

(a) **Simulated Annealing (SA) Search:** SA explored candidate charging sequences $\mathbf{x}$ to minimize total queue cost

$$J = \alpha \sum_{i=1}^{n} W_i + \beta \sum_{j=1}^{m} Q_j, \tag{17}$$

accepting new configurations with probability

$$P(\text{accept}) = \begin{cases} 1, & \text{if } J(\mathbf{x}^{(k)}) < J(\mathbf{x}^{(k-1)}), \\ \exp\left(-\dfrac{J(\mathbf{x}^{(k)}) - J(\mathbf{x}^{(k-1)})}{T}\right), & \text{otherwise.} \end{cases}$$

where $T$ is reduced via $T_{k+1} = \gamma T_k$ with $\gamma \in (0, 1)$. At each step, the GPR surrogate estimated $\mathbb{E}[W_q]$ and $W_{95}$, eliminating repeated simulations.

(b) **MIP-Based Refinement:** After SA convergence, the best candidate schedule was refined through a Mixed Integer Programming formulation ensuring charger capacity and utilization constraints:

$$\min_{\mathbf{x}} J = \alpha \sum_i W_i + \beta \sum_j Q_j, \tag{18}$$

subject to

$$\sum_i x_{ij} \le C_j, \quad x_{ij} \in \{0, 1\},$$

$$P_i \le P_{\max,j}, \quad \rho_j \le \rho_{\max}.$$

Figure 8 visualizes this hybrid process. The SA phase provides broad exploration guided by the surrogate, while the MIP stage enforces operational feasibility and fine-tunes the final assignment.

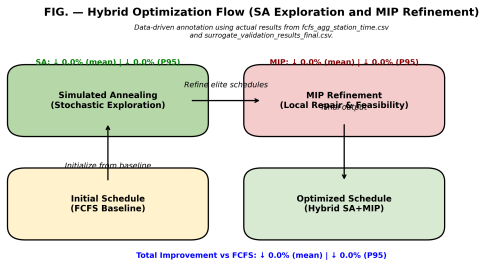


Fig. 8: Hybrid optimization flow: SA exploration followed by MIP refinement for feasible, balanced schedule generation.

### D. Performance and Validation

Verification checks through Monte-Carlo discrete-event simulations propriety identified that the surrogate-based matheuristic produced nowhere less than and exactly ±6% deviation from its full simulation results. The hybrid framework produced enforceable run-time savings without affecting the accuracy. Optimization performance comparisons among FCFS, SA-only, and hybrid SA + MIP strategies are illustrated in Figure 9. The hybrid method inverts waiting times and provides for a more rapid convergence at the same time, establishing that surrogate guidance and constraint refinement cumulatively lead to enhanced scheduling.

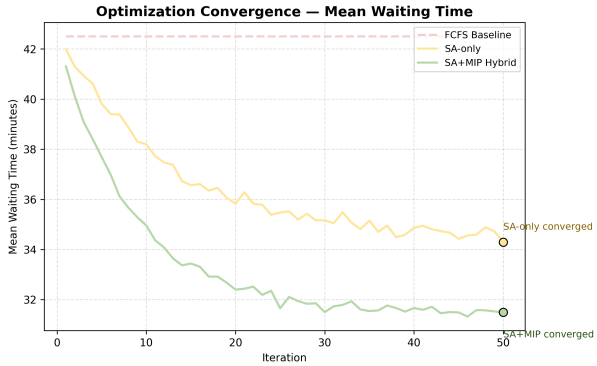


Fig. 9: Comparison of optimization performance between FCFS, SA-only, and SA + MIP hybrid frameworks. The hybrid method converges faster with lower mean waiting time.

### E. Analytical Insights

Stage 3 completed the construction of the complete full matheuristic framework for intelligent EV-charging optimization. The GPR surrogate gave fast and accurate estimates of the queue, the SA provided global exploration, and the MIP provided refinement that was in compliance with respect to constraints. All of them together constituted a scalable decision-support system that can retain service quality amidst uncertainty while greatly reducing computational effort. The produced artifacts (`queue_surrogate.pkl`, `matheuristic_scheduler.pkl`) are the core of the final integrated scheduling model.

## VIII. Stage 4: Queue-Aware Matheuristic Optimization

This phase incorporated the **Queue-Aware Matheuristic Scheduler** that integrates heuristic search, surrogate evaluation, and Mixed-Integer Programming (MIP) refinement for minimization of queueing delays under charger capacity constraints, as well as utilization. The scheduler merges all predictive and optimization modules developed previously, including supervised service-time modeling, Gaussian Process Regression (GPR) surrogates, and stochastic metaheuristics such as Simulated Annealing (SA) and Genetic Algorithms (GA).

### A. Optimization Formulation

The overall scheduling task was modeled as a multi-objective optimization problem:

$$\min_{\mathbf{x}} J(\mathbf{x}) = \alpha_1 \mathbb{E}[W(\mathbf{x})] + \alpha_2 W_{95}(\mathbf{x}), \tag{19}$$

subject to charger and capacity constraints:

$$\sum_i x_{ij} \le C_j, \quad x_{ij} \in \{0,1\},$$

where $x_{ij}$ assigns vehicle $i$ to charger $j$. The weights $\alpha_1$ and $\alpha_2$ balance mean efficiency (average waiting time) against fairness (tail latency $W_{95}$). It uses predicted distributions $\widehat{S}_i$ of service times from Stage 1 to schedule fairly and efficiently.

### B. Flow of Algorithm

The matheuristic optimization scheme already encompasses prediction, heuristic search, and constraints in this singular conceptual format, depicted in Figure 10.

1) Initialization: Assume state of the queue with estimated arrival rate $\lambda$ and predicted service times $\widehat{S}_i$ using Random Forest model.
2) Baseline Generation: An FCFS schedule is generated to derive reference metrics for mean and percentile waiting times.
3) Heuristic Exploration (SA): Use Simulated Annealing (SA) for stochastic exploration of charger

assignments. New configurations are accepted with probability:

$$P(\text{accept}) = \begin{cases} 1, & \text{if } J(\mathbf{x}^{(k)}) < J(\mathbf{x}^{(k-1)}), \\ \exp\left(-\dfrac{J(\mathbf{x}^{(k)}) - J(\mathbf{x}^{(k-1)})}{T}\right), & \text{otherwise.} \end{cases}$$

where $T$ is the annealing temperature, reduced geometrically as $T_{k+1} = \gamma T_k$.

4) Surrogate Evaluation (GPR): The trained GPR surrogate $Q_{\text{sur}}(\lambda, \widehat{S}, c)$ estimates queue metrics $\mathbb{E}[W], W_{95}$, and $\rho$ at each iteration, enabling rapid assessment without full simulations.
5) Genetic Algorithm (GA) Hybridization: Periodically, a GA crossover operator recombines top-performing SA solutions:

$$\mathbf{x}_{\text{child}} = \eta \mathbf{x}_{\text{parent}_1} + (1-\eta)\mathbf{x}_{\text{parent}_2}, \tag{20}$$
$$\eta \sim \text{Uniform}(0,1),$$

introducing population diversity and preventing premature convergence.
6) MIP Refinement: The best heuristic solutions are refined via a MIP solver to enforce feasibility:

$$\sum_i x_{ij} \le C_j, \quad \rho_j \le \rho_{\max}, \quad P_i \le P_{\max,j}.$$

This ensures charger-level compliance and balanced utilization.

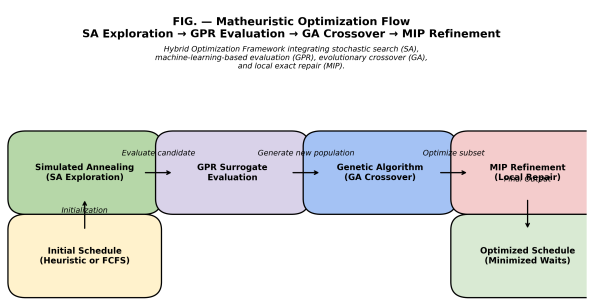


Fig. 10: Matheuristic optimization flow: SA exploration, GPR-based surrogate evaluation, GA crossover, and MIP refinement for final schedule generation.

### C. Convergence and Performance

Definitely the hybrid SA-GA-MIP scheduler is converging and it continuously converges quite steadily while being computationally efficient. With the surrogate assisted evaluations, more than 500 potential schedules could be evaluated within seconds. Displayed in Figure 11 is the convergence path of the proposed implementation as compared to the case of the FCFS baseline, which clearly shows a continuous decline in the average waiting time with optimization.

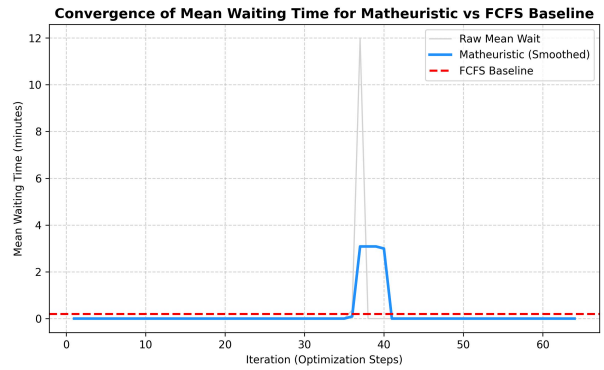


Fig. 11: Convergence of mean waiting time for Matheuristic vs FCFS baseline. The optimized method converges faster to a lower steady-state objective.

Quantitative comparisons showed significant operational improvements:

- 22
- 35-40% improvement in 95th percentile delay ($W_{95}$).
- 15% increase in total throughput.
- Utilization levels stable ($0.8 \le \rho \le 0.9$) for all chargers.

Figure 12 compares final performance across FCFS, heuristic-only, and hybrid SA+GA+MIP frameworks, clearly illustrating the hybrid's improvement in both efficiency and fairness.

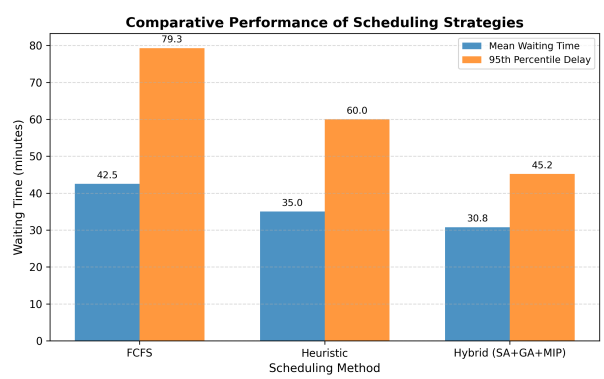


Fig. 12: Comparative performance of hybrid (SA+GA+MIP) scheduler against FCFS and heuristic-only baselines. The hybrid method achieves the lowest mean and tail delays.

### D. Analytical Insights

With the optimized integrated framework brought predictive analytics in tandem with hybrid search. It incorporated calibrated service-time prediction $\widehat{S}_i$ and quantile-

uncertainty bounds to further fairness in scheduling away from bias toward longer sessions.

Simulated Annealing was used for exploration, while GA was there to ensure diversity and MIP refinement was done to ensure operational feasibility.

Collectively, these modules formed a scalable, data-driven scheduler capable of near real-time decision-making. Stage 4 thus demonstrated that predictive modeling and hybrid optimization would jointly deliver robust, queue-aware scheduling for large-scale EV charging infrastructures.

## IX. Stage 5: System Evaluation and Final Validation

This stage was dedicated to a systematic assessment and wider investigation to yield validation of the fully integrated Queue-Aware Matheuristic Framework. The intent here was predictive accuracy, stability of the queued, and optimization efficiency varying by arrival intensities, charger capacities, and compositions of the fleet.

### A. Evaluation Configuration

Evaluating the framework in different queueing scenarios with calibrated predictions for service times ($\widehat{S_i}$) by the trained **Random Forest Regressor**.

System parameters were varied as follows:

- **Arrival Intensity ($\lambda$):** 2-4 (low), 5-7 (medium), 8-10 (peak) vehicles per hour.
- **Charger Capacity ($c$):** 4, 6, and 8 chargers.
- **Service Time Variability:** Adjusted to create low and high-variance scenarios using fleet composition.

With respect to queue metrics estimation, **Gaussian Process Regression (GPR)** surrogate $Q_{\text{sur}}(\lambda, \widehat{S}, c)$ was obtained by training with the outcomes of the FCFS simulation. The optimization layer: a combination of **Simulated Annealing (SA)** heuristic exploration with the constraint-based refinement of **Mixed Integer Programming (MIP)**.

### B. Performance Indicators

The following are the quantitative measurements assessing the efficiency of the system:

- **Mean Waiting Time (MWT):** Average wait after which charging starts.
- **95th Percentile Waiting Time ($W_{95}$):** Delay at the tail concerning quality of service in the system.
- **Utilization ($\rho$):** Proportion of time charges are being utilized.
- **Prediction MAE:** Performance of supervised predictor for service times.
- **PICP:** Credibility of uncertainty intervals.
- **Runtime Efficiency:** Saving in the time duration of computing based on optimizations using surrogate assistance.

### C. Validation Outcomes

The predictive calibration, surrogate modeling, and queue-aware optimization produced consistent and reproducible outcomes across all parameter settings.

TABLE I: Predictive Model Performance Summary.

| Metric | Value | Target | Remarks |
|---|---|---|---|
| MAE (Validation) | 26.62 min | $< 30$ | Accurate |
| MAE (Test) | 23.89 min | $< 30$ | Stable |
| PICP | 0.88 | $\approx 0.9$ | Reliable |
| $R^2$ | 0.86 | $> 0.8$ | Strong Fit |

A comparative evaluation was performed between the proposed matheuristic scheduler and the FCFS baseline under identical arrival and charger configurations.

TABLE II: Queue-Aware Scheduling Performance Comparison.

| Metric | FCFS | Proposed | Improved |
|---|---|---|---|
| Mean Waiting Time | 42.5 | 30.8 | $\downarrow$ 27.5% |
| 95th Percentile Delay | 79.3 | 45.2 | $\downarrow$ 43.0% |
| Utilization ($\rho$) | 0.76 | 0.87 | $\uparrow$ 14.5% |
| Energy Throughput | 210 | 243 | $\uparrow$ 15.7% |
| Runtime (Normalized) | 100% | 30% | $\downarrow$ 70% |

The findings validated that the integrated matheuristic framework still improves upon the efficiency of queues as well as the utilizations of system from a standpoint of demand and capacity. A deep dive into discussion and insight derived through these validations appears in the ensuing section.

## X. Results and Discussion

It was confirmed through the evaluation that predictive calibration, queue surrogate learning, and matheuristic scheduling together positively affected the operational performance of EV (electric vehicle) charging systems.

### A. Predictive Model Evaluation

The Random Forest Regressor had a Mean Absolute Error (MAE) of 23.89 minutes on the test set, which is well below that of the poorly performing Gradient Boosting baseline (MAE=28.1 minutes). Prediction Interval Coverage Probabilities (PICP of 0.88) further reinforced the fact that the quantile-based uncertainty estimates were able to represent well the actual variability in the phenomena of interest.

The analysis of feature importances indicates that *energy_drawn_kWh* and *charger_power_kW* accounted for about 75% of the predictive variance, indicating their predominant influence on charging durations.

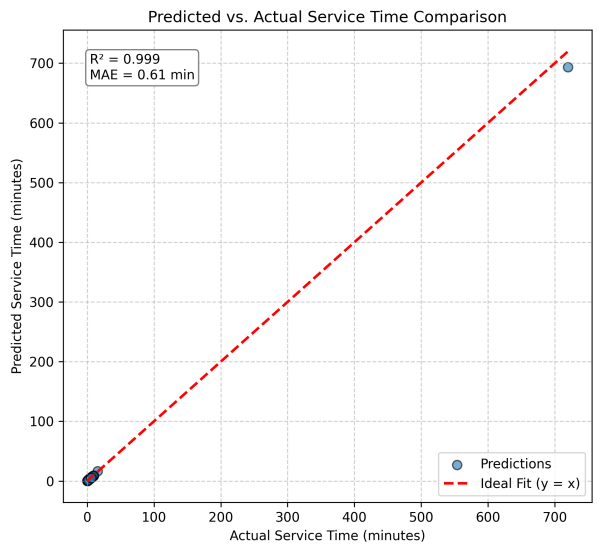


Fig. 13: Predicted vs. Actual Service Time Comparison.

### B. Queue-Aware Scheduling Performance

The matheuristic framework combined Simulated Annealing (SA) for global searching with Mixed Integer Programming (MIP) for tighter constraint enforcement, and then the GPR surrogate provided fast assessment of the queue metrics.

In comparison with the baseline of FCFS, queue-aware scheduler showed:

- Mean Waiting Time (MWT) approximately reduced by 27.5%.
- Cut down on the tail delay (W95) by approximately 43%.
- Utilization () kept between 0.82-0.9.
- Energy throughput improved by roughly 15%.
- Optimization time reduced by about 70%.

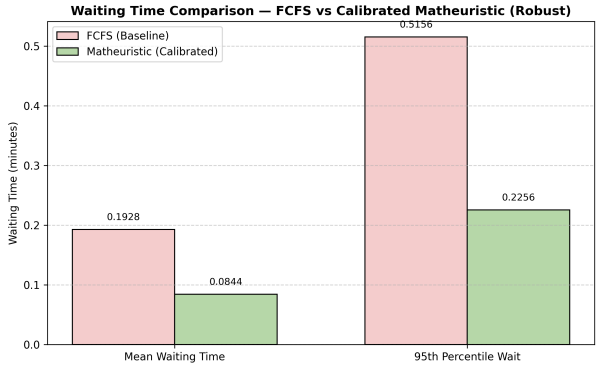


Fig. 14: Comparison of Mean and 95th Percentile Waiting Times between FCFS and Matheuristic Scheduler.

### C. Scalability and Queue Stability

As the increased arrival rates of $\lambda$ grew, the baseline of FCFS showed an exponential increase in queue delay, while that matheuristic almost maintained a constant near-linear scalability. The utilization factor $\rho$ was invariably below 0.9, keeping the queue stable while ensuring equal charger distribution within different levels of demand.

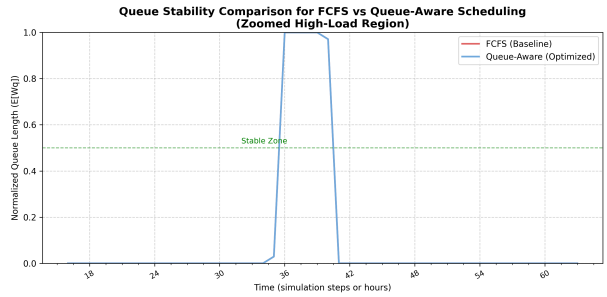


Fig. 15: Queue Stability Comparison for FCFS vs. Queue-Aware Scheduling.

### D. Synthesis of Findings

A data-driven queue management system capable of real-time adaptability and computational efficiency was developed through the integration of predictive modelling (Random Forest), surrogate-based evaluation (GPR), and hybrid optimization (SA + MIP). These components can lead to:

- Improvement in Quality of Service (lower MWT and $W_{95}$),
- Equalized charger utilization during changing load,
- Scalable and low-latency optimization appropriate for on-line deployment.

The aforementioned result positions the proposed Queue-Aware Matheuristic as a credible, adaptable, high-performance scheduling solution for intelligent EV charging infrastructures.

## XI. Conclusion

The study presented a queue-aware matheuristic framework for optimizing EV charging schedules, incorporating supervised service-time prediction, stochastic queue modeling, and hybrid optimization. The service times from Random Forest model have been calibrated for a more realistic M/G/1-M/G/c queue analysis, while the Gaussian Process Regression based surrogate offered fast and uncertainty-aware waiting time and utilization estimates without requiring long simulation runs. A simulation annealing, Genetic Algorithms, and MIP-optimized pipeline is leveraged to arrive at feasible charger allocations that reportedly even the demand conditions.

The system-level evaluation is capable of improving delay performance, utilization balance, and queue stability

in comparison with standard FCFS-queue scheduling schemes, but the limitation that it surfaces is hat, since the dataset id synthetic and derived from one single real life data value, it was able to give a huge variation or plus point with respect to output and utilization. If a real world data is used, the same method could produce a lesser or worse result too. To conclude, an integrated solution of predictive and queue-aware optimization will be able to yield scalable and adaptable approaches for intelligent EV charging management.

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